

We are proposing extending our multi-task optimizer research to transformer-based Large Language Models (LLMs) by conducting a comprehensive benchmark of optimization algorithms—including our novel MTMuon optimizer—across diverse NLP tasks and datasets. We will fine-tune shared LLM backbones (DistilBERT, RoBERTa, ALBERT) on multiple text classification and named entity recognition datasets spanning various domains and complexities, evaluating optimizers under controlled conditions with systematic variation in batch sizes, model architectures, and task pairings. The study aims to validate MTMuon's Nash equilibrium-based gradient balancing mechanism in NLP contexts and provide empirical insights into optimizer performance across different data characteristics.

The methodology employs a full factorial experimental design comparing seven optimizers (SGD, RMSprop, Adam, Muon, MTMuon, Adadelta, Adagrad,) across twelve datasets, three batch sizes, and three model architectures. Key outcomes will include performance rankings, convergence analyses, and practical guidelines for optimizer selection in multi-task NLP scenarios. The project will deliver an open-source benchmarking toolkit, detailed experimental results, and evidence-based recommendations, contributing to both optimization theory and practical LLM fine-tuning workflows while ensuring reproducibility through rigorous statistical analysis and comprehensive documentation.

Multi-task learning (MTL) improves model generalization by leveraging shared representations across related tasks. In our prior work, we introduced **MTMuon**, an optimizer that uses **equal task weighting (0.5/0.5)** inspired by Nash equilibrium to prevent task dominance in CNN-based vision models. The method demonstrated stable convergence and competitive performance without manual hyperparameter tuning.

To assess MTMuon's broader applicability, we now transition to **LLM fine-tuning**, where multi-task optimization is equally critical but understudied. This project will bridge the gap between vision and language domains, offering a principled evaluation of optimizer performance in transformer-based MTL.

Objectives

1. **Implement a multi-task LLM framework** with a shared transformer encoder and task-specific heads for classification and NER.
2. **Benchmark multiple optimizers** (SGD, RMSprop, Adam, adagrad, adadelta, Muon, MTMuon) across different batch sizes (8, 16, 32).
3. **Evaluate MTMuon's effectiveness** in balancing NLP tasks without manual weight tuning.
4. **Compare convergence speed, final accuracy, and F1 scores** across optimizers and architectures.

Model Architecture

- **Backbone:** DistilBERT or RoBERTa-base (shared encoder, frozen initially).
- **Task Heads:**
 - **Classification Head:** Linear layer on [CLS] token output.
 - **NER Head:** Linear layer on each token output (sequence labeling).

Datasets

Text Classification Datasets

Dataset	Domain	Classes	Samples	Key Characteristics
IMDb	Movie Reviews	2	50,000	Sentiment analysis, balanced
AG News	News Articles	4	120,000	Topic classification, long documents
20 Newsgroups	Discussion Forums	20	18,846	Fine-grained topics, technical vocabulary
DBPedia	Wikipedia	14	560,000	Ontology classification, large scale
TREC-6	Questions	6	5,452	Question type classification
Emotion	Social Media	6	20,000	Emotion detection, informal language

Named Entity Recognition Datasets

Dataset	Domain	Entity Types	Tokens	Key Characteristics
CoNLL-2003	News	4	300,000	Standard benchmark, clean text
BC5CDR	Biomedical	2	150,000	Scientific domain, specialized vocabulary
WNUT 2017	Social Media	6	70,000	Noisy text, emerging entities
OntoNotes 5.0	Mixed	18	2,000,000	Diverse domains, hierarchical entities
BioNLP 2013	Biomedical	13	450,000	Complex entity relations
MultiNERD	Multilingual	8	2,100,000	Cross-lingual transfer potential

Task Pairing Strategy: Task pairing is essential to prove MTMuon's generalizability and robustness across diverse scenarios, rather than just one specific case

1. **Pairing A:** IMDb (Classification) + CoNLL-2003 (NER) - Standard benchmark
2. **Pairing B:** AG News (Classification) + BC5CDR (NER) - News domain specialization
3. **Pairing C:** Emotion (Classification) + WNUT 2017 (NER) - Social media challenge

Without task pairing:

3 models × 7 optimizers × 3 batch sizes = 63 experiments

With task pairing:

3 models × 7 optimizers × 3 pairings × 3 batch sizes = 189 experiments

Multi-Task Training Strategy

- **Batch Routing:** Alternate between classification and NER batches.
- **Loss Balancing:**

MtMuon - $L_{\text{combined}} = (L_{\text{cls}} + L_{\text{ner}}) / 2$

- Baseline: Sum loss, uncertainty weighting, GradNorm, etc.

Experimental Setup

- **Batch Sizes:** 8, 16, 32
- **Epochs:** at least 6.
- **Metrics:**
 - Classification: Accuracy
 - NER: F1-Score (micro-averaged)
 - Convergence: Epochs to target performance
 - Efficiency: memory usage

Expected Outcomes & Deliverables

1. **Comparative performance tables** for all optimizers across batch sizes and tasks.
2. **Analysis of MTMuon's generalization** from vision to language models.
3. **Final report & presentation** detailing methodology, results, and insights.